

B. Tech. Degree VI Semester Examination April 2014**ME 602 DYNAMICS OF MACHINERY**
(2006 Scheme)

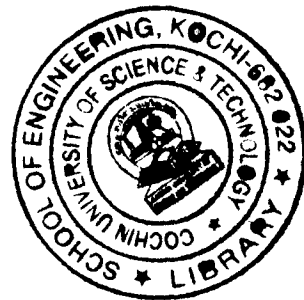
Time : 3 Hours

Maximum Marks : 100

PART A
(Answer ALL questions)

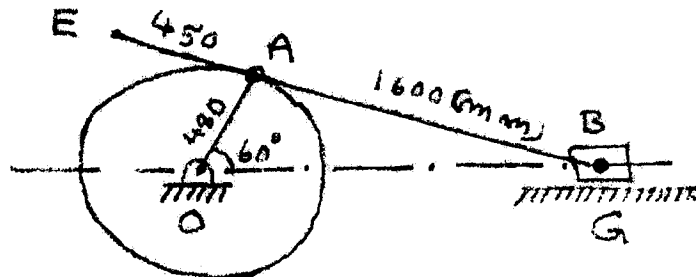
(8 x 5 = 40)

- I. (a) Sketch inversions of slider crank chain.
 (b) Compare cycloidal teeth and involute teeth forms of a gear.
 (c) Explain gyroscopic effect on ships.
 (d) What is hunting of governors?
 (e) Differentiate between static and dynamic balancing.
 (f) Explain the condition of complete balancing of an engine.
 (g) Explain the working principle of a rope brake dynamometer.
 (h) What are the advantages of V belt over flat belt?

**PART B**

(4 x 15 = 60)

II.



For the configuration of slider crank mechanism shown in figure, calculate.

- (i) The acceleration of slider at 'B'
 (ii) The acceleration of point E
 (iii) The acceleration of link A B

O A rotates at 20 rad/s counter-clockwise. All dimension are in mm.

OR

III.

The following data related to a horizontal reciprocating engine.

Mass of reciprocating parts	= 120 kg
Crank length	= 90 mm
Engine speed	= 600 rpm
Connecting rod mass	= 90 kg
Length between centers	= 450 mm
Distance of centre of mass from big end centre	= 180 mm
Radius of gyration about an axis through centre of mass	= 150 mm

Find the magnitude and direction of the inertia torque on the crank shaft when the crank has turned 30° from the inner dead-centre.

(P.T.O.)

- IV. A punching machine carries out 6 holes per minute. Each hole of 40mm diameter in 35mm thick plate requires 8Nm of Energy/mm² of sheared area. The punch has a stroke of 95mm. Find the power of the motor required if the mean speed of the flywheel is 20m/s. If the total fluctuation of speed is not to exceed 3% of the mean speed, determine the mass of the fly wheel

OR

- V. A porter governor, each of four arms is 400 mm long. The upper arms are pivoted on the axis of the sleeve where as the lower arms are attached to the sleeve at a distance of 45mm from the axis. Each ball has a mass of 8kg and the load on the sleeve is 60kg. What will be the equilibrium speeds for the two extreme radii of 250mm and 300mm of rotation of the governor balls?

- VI. Four masses A, B, C and D are completely balanced. Masses C and D makes angle of 90° and 195° respectively with B in the same sense. The rotating masses have following properties

$$m_b = 25\text{kg}, m_c = 40\text{kg}, m_d = 35\text{kg}$$

$$\sqrt{a} = 150\text{mm}, \sqrt{b} = 200\text{mm}, \sqrt{c} = 100\text{mm}$$

$$\sqrt{d} = 180\text{mm}$$

Planes B and C are 250mm apart.

Determine: (i) The mass A and its angular

(ii) The positions of planes A and D

OR

- VII. The cranks of a four cylinder marine oil engine are arranged at angular intervals of 90°. The engine speed is 70 rpm and reciprocating masses per cylinder is 800 kg. The inner cranks are 1m apart and are symmetrically arranged between the outer cranks which are 2.6m apart. Each crank is 400mm long. Determine the firing order of the cylinders for the best balance of reciprocating masses and also the magnitude of the unbalanced primary couple for that arrangement.

- VIII. A single plate clutch transmits 25kW at 900 rpm. The maximum pressure intensity between the plates is 85KN/m². The outer diameter of the plate is 360mm. Both the sides of the plate are effective and the coefficient of friction is 0.25. Determine: (i) The inner diameter of the plate (ii) The axial force to engage the clutch.

OR

- IX. In a belt transmission dynamometer, the driving pulley rotates at 300 rpm. The distance between the centers of the driving pulley and the dead mass is 800mm. The diameter of each of the driving as well as the intermediate pulleys is equal to 360mm. Find the value of the dead mass required to maintain the lever in a horizontal position when the power transmitted is 3kW. Also find its value when the belt just begins to slip on the driving pulley. μ being 0.25 and the maximum tension in the belt 1200N.
